

EDUCATION

University of Waterloo | Computer Engineering, BAsC.

Expected Graduation in April 2028

TECHNICAL SKILLS

Languages & Frameworks: C++ / C (MISRA), Python, Torch, ROS2, FreeRTOS, React, Angular, Node, FastAPI, Flask, Django, MongoDB, Unity/C#, MATLAB, platform.io, VHDL

Tools: Docker, Linux, Git, Slurm, CMake, KiCad, Onshape/Solidworks, COMSOL, Quartus Prime, Blender, ESP-IDF

Hardware: STM32, ESP32, NVIDIA Jetson, I2C, UART, CAN, DDS, FPGA, Saelae Logic Analyzer, Oscilloscope, Function Generator, Power Supply, Soldering

EXPERIENCE

Software Engineering Intern

Independent Robotics, Montreal, Canada

C++, Python, Docker, ROS2, PX4 Autopilot, Unity, GNSS RTK, AngularJS

May - August 2025

- Designed and deployed a Docker monorepo infrastructure and HIL testing pipeline for a Jetson & PX4 based autonomous boat, reducing feature development cycles by **40%**.
- Developed containerized **ROS2** nodes for manual and autonomous navigation, long-range serial-to-RF, embedded comms & sensor integration, TCP-Unity sim, and a corresponding Angular.js mission control dashboard.
- Authored a **GNSS RTK driver** with covariance estimation via Jacobian error propagation achieving **sub-degree attitude error**.
- Optimized a polar-to-Cartesian projection driver with Numpy for RTSP sonar, **trivializing latency** in real-time localization.

Software Engineering Intern

Rapyuta Robotics, Tokyo, Japan

C++/C, Python, Docker, ROS (1), Linux, KiCad, Onshape, Ultralytics, I2C/SPI, UART, ESP32

May - December 2024

- Designed a ROS **TOF** sensor crosstalk-aware obstacle filter FSM, **eliminating** robot collisions for weeks post-deployment.
- Developed a real-time on-device YOLOv8 perception POC, detecting misaligned ASRS bins in test with **95%** accuracy.
- Built and launched an internal-ops system monitoring device with touchscreen UI, Wifi & P2P packet monitoring, RFID reading for robot localization & I2C BMS interfacing.
- Prototyped, designed, soldered, and validated PCB, modeled form factors in Onshape, and programmed FreeRTOS ESP32 firmware. Deployed onsite, streamlining operations & earning **strong feedback** on usability and reliability.
- Developed a unified ROS diagnostic pipeline for all sensors and actuators on the main robot (Jetson) and battery exchange robot (RPI), exposing new error detection for all components and decreasing diagnostics **latency and load by 10x**.

Software Developer

WATonomous Student Design Team, University of Waterloo

C++/C, Python, ROS2, Torch, CUDA, TensorRT, Nuscenes, Anomalib

January 2025 - Present

- Developed end-to-end a Dockerized ROS2 differential drive robot navigation system in sim with costmap generation, map fusion, ray tracing, **A*** path planning and pure pursuit control .
- Built a Dockerized **camera & LIDAR** anomaly detection pipeline & trained an MSE Feature Reconstruction Error VAE
- Migrated YOLOv8-based autonomous car vision pipelines to **TensorRT** resulting in an **inference speedup of 200%**

Cancer Research Laboratory Intern

Dept. of Microbiology and Immunology, Western University

Laboratory of Rodney P. DeKoter

August 2021 - November 2022

- Used PCR & restriction enzyme methods to successfully clone EGFP and antibiotic resistance into a plasmid vector.
- Performed Chromatin Immunoprecipitation & Next Generation Sequencing, identifying transcription factor binding sites.
- Validated the viability of an engineered antibody through immunoblotting.
- Published in the Journal of Immunology (Impact Factor **5**), securing **\$50,000+** of grant funds. .

PROJECTS

Imaginecraft | Python, Torch, Gemini, KubeJS |

My Basement | August 2025

- Developed a serverside Minecraft mod with a custom backend that allows the actualization of a player's ideas directly into functioning game logic and items, through **MCP**-based asset and Javascript injection at runtime
- Designed and trained end-to-end a **custom CNN** and synthetic data pipeline through LLM assisted text-to-image diffusion and analyzed custom periodic loss functions to optimize rotationally symmetric coarse-to-fine semantic learning

Ross | Arduino, Python, OpenCV, SAM (Meta) |

Hack the North 2025, University of Waterloo | September 2025

- Built a system that transforms visual art into touch and sound for the blind. Selected as a **Finalist Project** out of **260+** teams.
- Engineered a 3DOF delta plotter using NEMA steppers, servos, 3D-printed parts, & finetuned GRBL software for arduino
- Built a pipeline combining Meta's SAM segmentation, graph-based stroke optimization, and Cohere+ElevenLabs narration to deliver synchronized tactile and auditory feedback.